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RESEARCH ARTICLE

A Multi-Agent System for Cybersecurity Threat Detection and Correlation Using Large Language Models

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ABSTRACT As cyber-attacks rapidly evolve across communication, infrastructure and data layers, traditional security solutions such as rule-based intrusion detection systems (IDS) or signature-based antivirus programs are effective at detecting known threats, but they often lack the contextual understanding and semantic interpretation necessary to detect complex or evolving attacks. For example, spear-phishing campaigns, advanced persistent threats (APTs), and multi-stage attacks often escape detection due to their subtle and context-dependent nature. This limitation creates a critical gap in detecting coordinated or subtle attack patterns that span multiple systems and domains. The need for semantic understanding, cross-domain visibility, and adaptive detection is increasingly urgent, particularly as threat actors employ polymorphic and AI-driven strategies that traditional systems cannot interpret or correlate effectively. This paper presents a modular multi-agent architecture that integrates established cybersecurity analysis tools with large language models (LLMs) to achieve intelligent, explicable and highly accurate detection of threats across diverse data types. Three specialized agents: 1) email verification, 2) log analysis, and 3) IP address scanning each operate independently with tailored detection pipelines that combine domain-specific tools and LLM-powered semantic analysis components to identify, characterize, and report threats specific to their domain. At the core of the system lies a contextual recommendation system that processes and cross-analyzes the outputs of all specialized agents to detect complex threat patterns such as multi-vector, time-based, or stealth attacks that would otherwise evade isolated detection mechanisms. The evaluation on benchmark datasets, including CIC-IDS 2017, SpamAssassin, and custom simulated network environments, demonstrates threat detection accuracy of 93.6%, multi-agent correlation accuracy of 87%, and false positive reduction of 41.3% compared to traditional approaches. The use of LLMs for both structured explanations and chain-of-thought reporting further enhances analyst confidence and reduces triage time.

INDEX TERMS Multi-agent systems, LLMs, contextual threat analysis, semantic analysis, email phishing detection, log-based anomaly detection, IP scanning.

LIST OF ABBREVIATIONS

AI	Artificial Intelligence.
API	Application Programming Interface.
CIDR	Classless Inter-Domain Routing.
CIC-IDS	Canadian Institute for Cybersecurity - Intrusion Detection System.
CVE	Common Vulnerabilities and Exposures.
CVSS	Common Vulnerability Scoring System.
DNS	Domain Name System.
ELK Stack	Elasticsearch, Logstash, Kibana
FAISS	Facebook AI Similarity Search.
FPR	False Positive Rate.
GPT	Generative Pre-trained Transformer.
IP	Internet Protocol.
LLM	Large Language Model.
Nmap	Network Mapper.
NVD	National Vulnerability Database.

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OSINT	Open Source Intelligence.
RAG	Retrieval-Augmented Generation.
RegEx	Regular Expression.
SIEM	Security Information and Event Management.
SMTP	Simple Mail Transfer Protocol.
SOC	Security Operations Center.
SSH	Secure Shell.
Suricata	Open-source Network Threat Detection Engine.
TLD	Top-Level Domain.
URL	Uniform Resource Locator.
XAI	Explainable Artificial Intelligence.

I. INTRODUCTION

The sophistication and frequency of cyberattacks continues to grow, with increasingly interconnected digital systems targeted. Whether socially engineered phishing campaigns or sneaky lateral movements across compromised endpoints, modern threats are rarely isolated. They cover multiple vectors such as email, logs and IP communications, thus creating detection challenges that exceed the capabilities of siloed or rule-based security systems. Conventional signature and model-based approaches, despite their effectiveness against known threats, often fail to detect latent indicators distributed across different domains, limiting their ability to support timely and explainable decision-making [1], [8]. To remedy these limitations, researchers have looked to multi-agent architectures in the field of cybersecurity. Such systems decentralize detection logic into domain-specific intelligent agents, each one responsible for analyzing a particular threat layer [43]. This modular design provides scalability, resilience and parallel processing capabilities [6], [11]. However, many of these frameworks still suffer from a lack of deep semantic reasoning or natural language understanding, making them hard to interpret and limited in terms of adaptability when dealing with polymorphic or AI-driven threats [25], [27].

The emergence in recent years of LLMs such as GPT-4 and LLaMA has created new avenues for the application of natural language processing and contextual interpretation to cybersecurity problems [40], [44]. LLMs have shown strong capabilities in interpreting unstructured data, generating threat descriptions and even in interacting with live security environments through chain-of-thought and retrieval-augmented generation [2], [4], [17]. Their effectiveness in tasks varying from phishing detection to network protocol analysis has been demonstrated in studies [3], [19], [20]. Although LLMs have shown strong capabilities across various cybersecurity tasks, their integration into multi-agent security systems for correlating diverse threat indicators and producing interpretable outputs is still at an early stage of development [16], [18].

In this work, we present an LLM-enhanced multi-agent cybersecurity system designed to detect, correlate and interpret cyberthreats in three main areas: i) e-mail traffic, ii) server logs and iii) IP range activities. The architecture includes three intelligent agents, each equipped with classic tools, combined with refined LLM prompts that enhance semantic understanding [21], [29]. Each agent operates autonomously and generates structured results and risk signals that are sent to a central recommendation system component. This system contextualizes the indicators, correlates threats between domains and synthesizes a final threat description [11], [31].

Our system is evaluated on real and synthetic datasets [12], including SpamAssassin corpus, LLM-generated phishing samples, CIC-IDS 2017 [13] logs and emulated IP scanning activities. The system achieves 93.6% detection accuracy, 87% correlation accuracy and a 41.3% reduction in false positives compared to traditional rule-based or single-agent pipelines. LLM-based incorporation of explicability also demonstrated a significant increase in analyst confidence and a decrease in triage time in simulated red-team scenarios [23], [36].

Our approach is based on four contributions. Firstly, we propose a modular multi-agent architecture for cyberthreat analysis enhanced by LLM-based reasoning [5], [16]. Secondly, we introduce a contextual recommendation system correlating inter-agent proofs into coherent narratives [10], [37]. Thirdly, we show a series of prompt-based pipelines that allow for explainable results from agents [28], [39]. Finally, we evaluate the entire system using both public and synthetic datasets underlining its operational impact and scalability [7], [35].

The paper is organized as follows. Section **II** provides a review of existing research on multi-agent cybersecurity systems, LLM applications and semantic threat correlation. Section **III** presents the architecture of our system, including the agent pipelines and the recommendation system. Section **IV** details the implementation and integration of the tools. Section **V** discusses the results of our experiments. Section **VI** sets out future directions and concludes the paper.

II. BACKGROUND

The growing complexity of cyber threats has sparked extensive research into enhancing detection, correlation, and interpretability within cybersecurity systems. Scholars have explored multiple interdependent avenues as shown on Table 1: the decentralization of detection through multi-agent architectures [1], the integration of LLMs into security pipelines [2], the evolution of phishing detection via AI [3], sophisticated contextual threat correlation [4], and explainability frameworks for trust and usability [5], [34]. These threads form the foundation upon which our system builds, linking LLM capabilities to a modular, workflow-driven approach rather than cognitively autonomous agents.

Decentralized systems have emerged as resilient strategies for cyber threat response in dynamic environments. Soltani et al. [6] introduced a multi-agent deep learning framework enabling agents to adaptively detect intrusions in distributed networks, while Liu [11] proposed LLM-based

TABLE 1. Comparative Analysis of multi-agent systems.

Feature / System	LLM Integration	Multi-Agent Design	Threat Correlation	Explainability (XAI)	Domain Coverage	Advantages	Disadvantages
Afane et al. [1]	Yes (adversarial use of LLMs)	LLM agents used for phishing generation)	No	No	Phishing Emails	Demonstrates generative use of LLMs for phishing scenarios	Lacks detection or correlation capabilities
Koide et al. [3]	Yes (GPT-4 Vision)	No	No	Strong visual-based XAI	Web-based phishing	Provides visual interpretability using advanced vision-based LLMs	No agent design or multi-domain correlation
Soltani et al. [6]	No	Yes (adaptive deep learning agents	Partial (per-agent learning)	No	Network traffic	Adaptive DL-based agentic approach	No semantic correlation or centralized control
Kasri et al. [5]	Yes	No	Theoretical only	Conceptual focus on LLM benefits	General cybersecurity	Discusses theoretical LLM benefits in security	No practical implementation or empirical validation
Shafee et al. [10]	Yes (LLM-based OSINT bots)	No	No	Yes (chat-based explanations)	Threat Intelligence (OSINT)	Implements chat-driven explainable OSINT tool	Lack of multi-domain threat correlation
Xu et al. [25]	No	Yes (Edge-AI agents)	Yes (cooperative edge model)	Yes (game-based, interpretable)	Advanced Persistent Threats (APTs)	Edge-ready agentic structure with collaborative correlation	High complexity and hardware dependency
Yigit et al. [32]	Yes (critical infrastructure XAI)	Yes (distributed agentic security)	Yes (benchmarking and correlation)	Yes (explainable audit logs)	Critical Infrastructure Protection	Fully explainable, distributed, and correlated security pipeline	Complex deployment and scaling challenges
Marantos et al. [37]	Yes (scenario simulation and threat modeling)	Yes (agent coordination for training)	No (focus on training realism)	Yes (scenario feedback)	Security Training	Ideal for training use cases with realistic scenario feedback	Not focused on operational detection or correlation

collaboration for incident response. Hasanov et al. [17] underscored multi-agent designs as scalable paradigms for cyber defense. Beyond academic conceptualizations, industry initiatives such as those explored by Kshetri [33] suggest that agentic AI will increasingly underpin critical infrastructure defense. Nonetheless, these systems often lack semantic coherence between detection components, prompting interest in workflow-based models like ours that maintain interpretability without full agent autonomy.

In parallel, LLMs have rapidly infiltrated cybersecurity workflows, powering log analysis, threat summarization, and attack simulation. Kasri et al. [5] provided a systematic assessment of LLMs for tasks ranging from anomaly detection to malware classification. Liu et al. [4] highlighted their flexibility in protocol analysis under resource constraints, while Chen et al. [18] and Guven [24] addressed both operational advantages and security risks. Recent work has expanded this view: Yigit et al. [32] proposed hybrid LLM architectures for infrastructure protection, and Braun [36] evaluated methodologies for explainability in agent-LM coordination. Such efforts illustrate the range and versatility of LLMs, yet also support our stance that these models are most effective when structured within strict pipelines rather than autonomous entities.

Phishing detection, as a historically dominant attack vector, remains a central use case for LLMs in cybersecurity. Koide et al. [3] introduced ChatPhishDetector, capable of analyzing multilingual phishing threats using GPT-4 Vision. Afane et al. [1] revealed how attackers exploit LLMs to craft adaptive, evasive phishing content, emphasizing the escalating arms race. Altwaijry et al. [19] and Atawneh and Aljehani [20] benchmarked deep learning-based detectors, while Trad and Chehab [23] explored LLM prompt engineering versus fine-tuning for this task. More recently, López Delgado and López Ramos [41] have examined phishing detection pipelines that leverage soft attention mechanisms, pointing to novel integration paths for natural language processing and security signal recognition.

To go beyond detection, a growing body of work now addresses how to correlate disparate threat signals across environments. For example, Landauer et al. [8] surveyed anomaly detection techniques in log sequences, while Ruzickova et al. [9] proposed narrative construction through LLM-based post-event synthesis. Approaches such as LogPrompt by Liu et al. [21] facilitate zero-shot log interpretation, and Lohar and Baraskar [22] implement a complete LLM-driven pipeline from log ingestion to actionable alerts. Additionally, Marantos et al. [37] presented a multi-layer correlation approach incorporating explainable AI for IoT security, reaffirming the importance of interpretability alongside automation. These studies support our strategy of linking security components—logs, network events, and email flows—through workflow-aware LLM modules to enable transparent cross-domain reasoning.

Explainability plays a pivotal role in the real-world adoption of AI-driven security systems. Balogh et al. [2] investigated how generative models enhance analyst comprehension, while Combs et al. [7] examined LLM reasoning through analogical lenses. Shenoy and Mbaziira [29] proposed frameworks for prompt engineering that align with analyst needs, and Garde et al. [30] offered techniques for summarizing posture indicators from multiple inputs. Notably, Pasca et al. [39] demonstrated that interpretable explanations significantly improve security decision-making, supporting our commitment to traceability and auditability in every step of our system.

Building upon these foundations, our system proposes a state-of-the-art architecture that leverages LLMs not as autonomous agents, but as modular processors integrated into a robust, workflow-driven cybersecurity pipeline. Each LLM module executes specific, well-defined tasks such as log summarization, phishing detection, or contextual linking, without engaging in independent reasoning [10], [16]. This design maximizes interpretability, reduces inter-module inconsistency, and facilitates traceable, multi-source correlation offering a scalable, secure, and auditable framework aligned with the critical needs of modern cyber defense [27], [31], [42].

III. SYSTEM ARCHITECTURE

Our proposed architecture presents a modular, workflow-oriented cybersecurity system designed to enhance threat detection, correlation, and interpretability across multiple data domains. The architecture focuses on practical integration between semantic analysis and structured decision-making while maintaining simplicity of deployment and transparency of operation. As illustrated in Fig.1, the system consists of four main layers: user interaction, a task dispatching mechanism, three specialized processing agents (for email, logs, and IP addresses), and a centralized cross-context recommendation system.

The process starts when a user submits one of three types of tasks: (i) email verification, (ii) security log analysis, or (iii) IP address range scanning.These tasks are received and routed by the task dispatcher agent, which ensures that each request is directed to the appropriate processing unit. Each of the three processing agents, email verification agent, log analyzer agent, and IP address range analyzer agent executes a predefined workflow that combines three core components: domain-specific tools (e.g., regex filters, DNS checks, anomaly scorers), a large language model (LLM) for semantic enhancement and pattern interpretation, and a lightweight memory module for intermediate storage and contextual continuity.

Each agent independently processes the input it receives, performs risk assessment, and calculates a final score or structured report describing detected threats. These intermediate outputs are sent to the cross-context recommendation system, a central analysis component responsible for synthesizing signals across the agents. For example, the system may detect a phishing email linked to an anomalous login timestamp and a suspicious IP address signals that, when considered in isolation, may appear benign. This semantic fusion enables high-confidence threat correlation and greatly reduces false positives.

Unlike systems that require tightly coupled, complex inference models, our architecture embraces simplicity and modularity. Each agent is designed to follow a clearly defined and auditable workflow, ensuring that outputs are consistent, traceable, and interpretable by both automated systems and human analysts. This not only facilitates debugging and tuning but also simplifies upgrades and the integration of new detection capabilities.

Fig. 1 captures the complete flow: user-defined tasks pass through the dispatcher, are analyzed in their respective domains, and the resulting insights are unified by the recommendation layer to produce a comprehensive threat assessment. The architecture, through its combination of semantic precision, domain specialization, and centralized correlation makes it an effective and practical choice for adaptive, explainable, and efficient cyber threat detection.

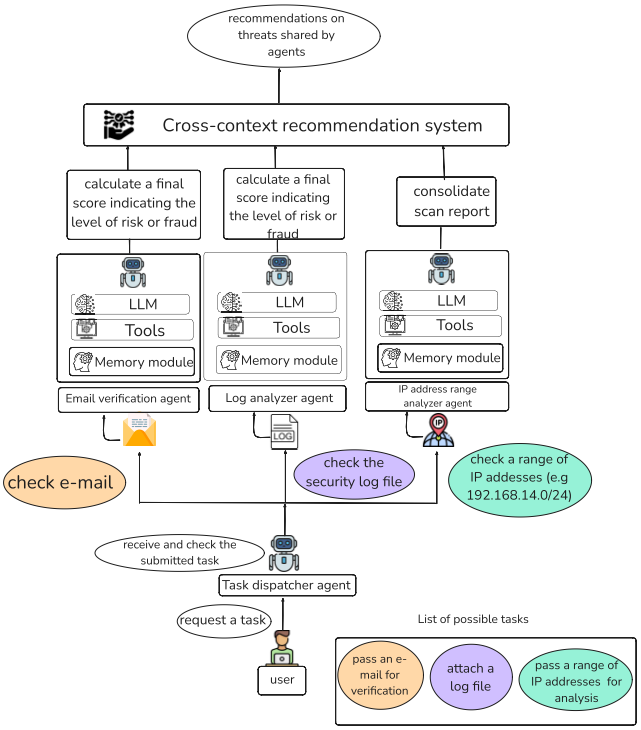


FIGURE 1. Overview of the global architecture.

A. TASK DISPATCHER AGENT

The task dispatcher agent serves as the central entry point and orchestrator for all user-submitted operations within the system. As shown in Fig.2, this agent manages three critical responsibilities: verifying input integrity, interpreting the type of cybersecurity analysis requested, and routing tasks to the appropriate domain-specific processing agent. This

agent plays a pivotal role in maintaining system reliability, extensibility, and responsiveness.

Upon receiving a request, the dispatcher first performs input validation to ensure that the submitted data whether an email, a log file, or a range of IP addresses is syntactically and structurally correct. The dispatcher agent also checks for completeness, such as ensuring the presence of metadata fields, proper formatting, and necessary contextual information. This verification stage prevents malformed or insufficient data from propagating through the analytical layers.

The dispatcher then determines the nature of the task: for instance, distinguishing whether the submission is intended for phishing verification, log anomaly analysis, or IP address reputation scoring. This classification is essential, as it triggers the activation of specialized pipelines with distinct tools, prompt strategies, and scoring mechanisms.

Once the task type is identified, the dispatcher securely routes it to the corresponding domain agent, the email verification agent, log analyzer agent, or IP address range agent. This design separates data ingestion from processing, ensuring that each analytical agent can remain optimized and focused on a single type of cybersecurity signal.

Moreover, the dispatcher manages asynchronous task handling, enabling parallel processing and scalable task queuing. This means the architecture can gracefully accommodate future expansions, such as integrating new agents (e.g., browser behavior trackers or mobile application monitors), without modifying the core logic of the dispatcher. The modularity and abstraction layer contribute significantly to the evolutivity and maintainability of the system.

By abstracting preprocessing, task classification, and routing into a unified gateway, the task dispatcher agent enforces discipline across the system, reduces error propagation and provides a clear control point for auditing and operational logging. The agent is the backbone that upholds the modularity and effectiveness of the entire architecture.

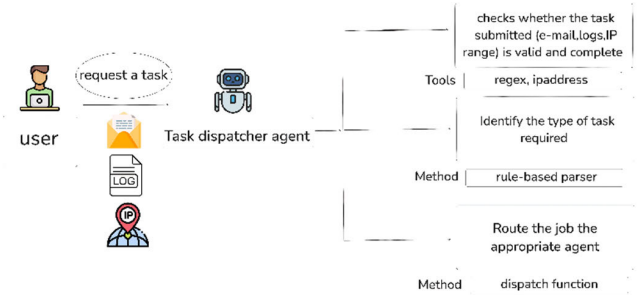


FIGURE 2. The detailed workflow of the task dispatcher agent.

B. EMAIL VERIFICATION AGENT

The Email Verification Agent is designed to identify a wide range of threats in email communications, from traditional phishing and spoofing to polymorphic and AI-generated attacks. As shown in Fig.3, this agent processes each submitted email through a carefully orchestrated workflow that integrates lightweight rule-based tools with advanced semantic analysis powered by LLMs.

Upon receiving a job from the dispatcher, the agent performs a series of pre-processing checks, extracting metadata such as the sender domain, message headers, and embedded URLs. This initial layer ensures structural validity and provides essential indicators for risk classification.

To identify content that has been reused or subtle variations in known phishing attacks, the agent uses a vector similarity search using FAISS. Each submitted e-mail is coded and compared with a pre-indexed vector database built from annotated datasets of legitimate and fraudulent e-mails. This approach enables rapid approximate matching, which is particularly useful for identifying threats that have been paraphrased or obscured by style.

At the same time, the agent applies symbolic analysis using tools such as RegEx (to spot keywords, brand spoofing and suspicious entities), tldextract and dns.resolver to decompose and validate URLs. These modular tools provide deterministic information, notably by analyzing links and identifying unusual domain behavior.

A distinguishing feature of the agent is its hybrid semantic reasoning layer, implemented using a Retrieval-Augmented Generation (RAG) architecture powered by LLaMA 3.3-70B Versatile through the Groq API. The RAG mechanism dynamically retrieves relevant contextual excerpts—such as similar phishing samples, behavioral patterns, or vector store indicators—and integrates them into a tailored prompt to guide the reasoning of the LLM. This approach ensures that the model's responses are grounded in known threat behaviors while maintaining the flexibility to interpret emerging indicators.

The LLM is responsible for synthesizing all intermediate results, technical signals, metadata anomalies, sentiment changes and vector matches into a natural language report. This report serves two purposes: (1) it provides a clear risk verdict (e.g., safe, suspicious, phishing) and (2) it provides a justification trail, helping analysts to quickly understand the reasoning behind each classification.

This design achieves a balance between performance, cost-effectiveness and transparency. The use of groq's ultra-fast inference API significantly reduces latency and computational overhead compared to local deployments, making the agent responsive enough for real-time applications. The modular nature of the tools also means that upgrades and recycles can be carried out without affecting the entire pipeline.

The email verification agent is a scalable, explainable and future-proof component that captures both surface and deep contextual signals. The layered approach is particularly effective against evolving phishing strategies, including those generated by adversary LLMs, and forms a central pillar of the domain-specific defense capability of the system.

The design of the email verification agent is based on recent advances in phishing email detection. Alhuzali et al. [52] conducted an exhaustive comparative analysis of fourteen machine learning and deep learning

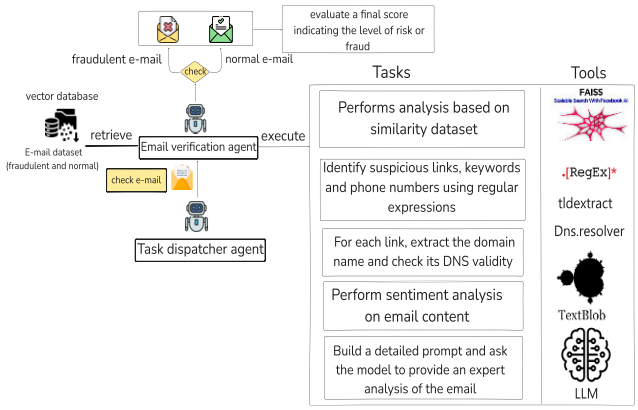


FIGURE 3. The detailed workflow of the email agent.

models, including BERT and LSTM, on ten publicly available datasets. Their study confirmed that SpamAssassin remains a relevant benchmark, despite its age, particularly when combined with modern architectures. In addition, Atawneh and Aljehani [20] demonstrated the effectiveness of deep learning models such as CNNs, RNNs, and BERT-LSTM, achieving accuracy of up to 99.61% on email datasets from public sources. These results support our approach of combining heuristic feature extraction with model-guided reasoning to improve the detection performance and explainability of the email verification agent.

C. LOG ANALYZER AGENT

The Log Analyzer agent is a crucial system component, as shown in Fig.4 designed to process structured system and network log files to detect malicious behavior that often escapes perimeter-based detection. These include patterns related to privilege escalation, lateral movement, persistence mechanisms and insider abuse, techniques common in advanced persistent threats (APTs). As shown in Fig.4, the agent which is based on a multi-stage process that combines deterministic filtering, rule-based inspection and contextual interpretation using a large language model (LLM), offers an alternative to traditional log analysis.

The analysis pipeline begins with pre-processing and normalization. Logs from various sources, such as system authentication logs (auth.log), web server access logs and IDS event logs, are first analyzed and normalized using the ELK stack (Elasticsearch, Logstash and Kibana). This ensures that all entries, regardless of their native format, are converted into a uniform schema. At this stage, irrelevant or redundant entries are filtered out, and fields such as timestamps, IP addresses and user IDs are extracted for subsequent use.

Once normalized, the logs are processed by a signature-based detection system powered by Suricata, a high-performance intrusion detection and prevention system. Suricata uses well-maintained community and proprietary rule sets to detect signatures of known threats such as SSH brute force attempts, abnormal DNS queries or port scanning behavior. This step enables rapid detection of known attack patterns with high accuracy, even if it may miss new or stealthy attack variants that don't match the predefined rules.

To remedy this limitation, the agent invokes its third, most innovative layer: contextual interpretation using an LLM, in particular LLaMA 3.3-70B accessible via the Groq API. The LLM receives sequences of suspicious log entries enriched with metadata such as host names, timestamps, event types and previous alert scores. The LLM follows a predefined prompt structure, and is responsible for summarizing events in a coherent, human-readable language, identifying abnormal patterns or escalation attempts, such as a successful sudo command after several failed authentications, deducing high-level tactics, such as recognition or persistence, based on time, user behavior and command types and providing confidence scores or risk levels, which feed the recommendation system.

This LLM integration adds semantic insight to the analysis. LLM excels in detecting soft signals, such as order usage or temporal anomalies, which are difficult to encode in strict rules. The ability of LLM to translate raw log sequences into a structured explanation considerably reduces the cognitive load of analysts. Each summary generated is traceable, referencing the exact lines of the log and preserving transparency, a crucial requirement of explainable AI (XAI) for cybersecurity.

In addition, the agent features a memory module for storing the results of recent analyses and high-risk patterns. This enables the agent to compare current logs with historical behaviors such as repeated attempts by the same user to access sensitive directories during different sessions, promoting temporal correlation and tracking the evolution of threats.

The log analyzer agent illustrates the synergy between structured rules systems and semantic LLM interpretation. This agent enhances threat visibility across multiple log sources, reinforces traditional signature detection and delivers detailed, interpretable results that help security teams make rapid, informed decisions. The use of a workflow-linked LLM instead of a free reasoning model ensures reliability, repeatability and ease of integration into real-world operations.

The implementation of the Log Analyzer agent is based on recent research on system log analysis and anomaly detection using deep learning. Le and Zhang [53] provided a comprehensive review of deep learning techniques applied to system logs, highlighting both their effectiveness and limitations in real-world environments. Their study focused on the challenges of generalization, false positives, and log diversity, which our agent addresses through modular reasoning and correlation layers. In addition, Xie et al. [54] presented LogGD, a graph neural network-based approach capable of capturing structural dependencies in log sequences. Their work demonstrated the importance of integrating contextual relationships between logs for accurate anomaly detection. These results confirm our architectural choices, which favor hybrid symbolic-neural reasoning over purely statistical classifiers for anomaly detection in logs.

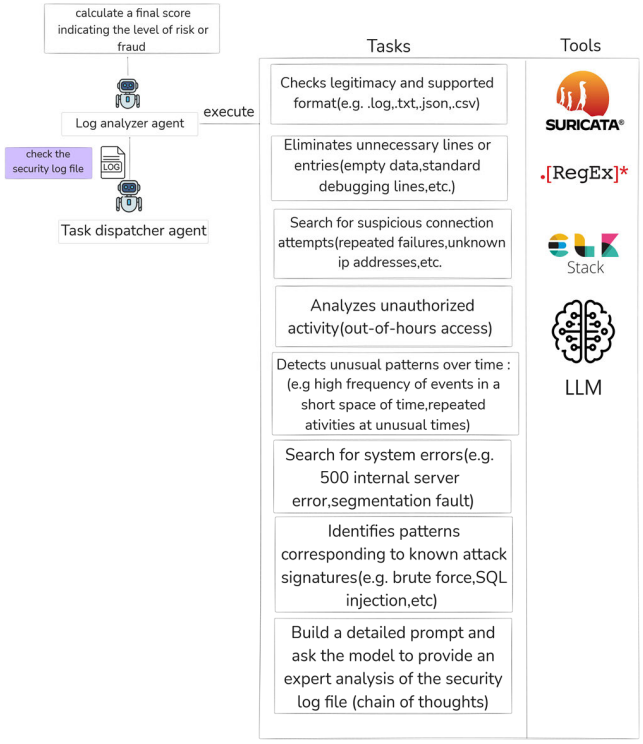


FIGURE 4. The detailed workflow of the log analyzer agent.

D. IP RANGE ANALYZER AGENT

The IP range analyzer agent is responsible for evaluating the exposure and vulnerability of a given set of networked hosts. Designed to process user-submitted IP address ranges (for example, in CIDR format such as 192.168.14.0/24), this agent automates the discovery of active nodes, services and associated risks using both innovative conventional tools and advanced semantic synthesis techniques. The complete workflow is illustrated in Fig.5.

Once a task is submitted, the agent first validates the input using the python module ipaddress, ensuring that it conforms to the correct IP address or CIDR syntax. The Agent then performs an in-depth analysis of the network via Nmap, identifying open ports and running services within the specified address range. This enables the agent to enumerate the available network surface and extract rich metadata about the detected hosts.

The next step is to go over the results of the analysis to generate a structured report describing each host, its accessible services and port configurations. For each service identified, version traces are compared with the National Vulnerability Database (NVD) [14] to extract known common vulnerabilities and exposures (CVEs). This includes CVSS scores, risk vectors and potential exploitability. These queries transform raw service data into actionable vulnerability information.

At the heart of this agent is its LLM-based analysis core, built on LLaMA 3.3-70B and accessible via the groq api. The reasoning is not delegated to the model, but follows a structured, deterministic workflow. The agent uses the analysis results to generate a well-formatted prompt that contextualizes the IP range, exposed services and mapped CVEs. The role of the LLM is to explain the risk position in natural language, converting the technical results of the analysis into accessible descriptions, classify host risk levels according to severity and type of exposed vulnerabilities, suggest prioritization strategies (for example, ''critical vulnerability on an outbound SSH port on host 192.168.14.3'') and tailor remediation advice to operational roles (security engineer or risk manager).

This workflow-driven design provides interpretation and control, avoiding the unpredictability of free-form generative models. The result returned to the task sender is a structured semantic summary that merges analysis data, CVE results and LLM-enhanced insights into a coherent, explainable safety report.

As shown in Fig.5, this agent illustrates a strong integration between deterministic scanning tools (e.g., Nmap, NVD API) and semantic augmentation via LLMs, offering cybersecurity operational teams not only raw alerts, but also contextual narratives on which they can act immediately.

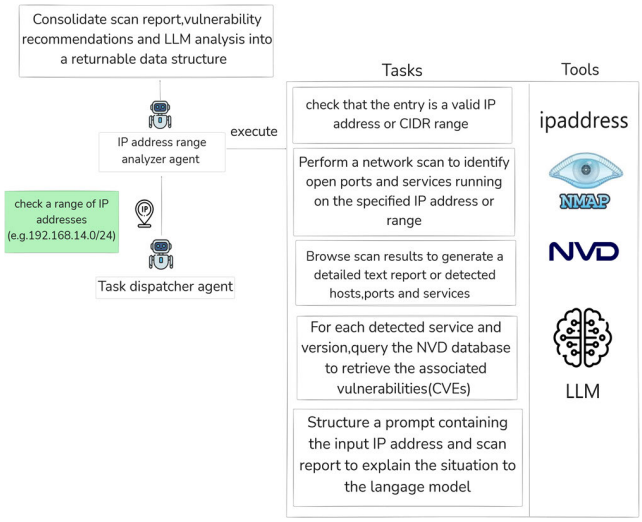


FIGURE 5. The detailed workflow of the IP range analyzer agent.

The IP Range Analyzer agent follows modern approaches to network recognition detection by using structured threat analysis and assessment techniques. Pittman's systematic study [55] emphasizes that port scanning is widely recognized as the first phase of cyberattacks and highlights the growing use of machine learning techniques for early detection of scanning behavior. Our agent uses Nmap for active network enumeration, enhanced by contextual reasoning using vulnerability metadata from the National Vulnerability Database (NVD). This hybrid approach, which combines active scanning and CVE semantic analysis, follows the logic of intelligent detection systems capable of quickly and explainably assessing exposure risks.

E. CONTEXTUAL RECOMMENDATION SYSTEM

The contextual recommendation system acts as the consolidation and correlation center of the architecture, responsible for

transforming agent-specific analytical results into actionable information about multi-domain threats. This system collects structured results from the email verification agent, log analysis agent, and IP range analysis agent. The system is designed to provide a comprehensive view of the threat landscape by combining data from multiple sources and integrating it into a single view.

When the system receives reports from the analytics agents, the first step is to normalize and aggregate the data, resulting in a consistent structure for further reasoning. Next, a search for matches is performed using regular expressions to detect common entities such as IP addresses, URLs, or domain names that may appear in different analysis contexts. For example, if an email analysis flags a suspicious domain and that same domain appears in a failed login event or an exposed service from an analyzed host, the system flags it as a correlated anomaly.

Beyond syntactic matches, the system also applies temporal correlation by examining the timing of events, identifying coordinated activities such as phishing campaigns followed by lateral movement. When multiple indicators confirm the link, the system assigns a correlation confidence level, indicating the likelihood that these events are part of a unified threat scenario.

To make these results understandable to analysts, the contextual recommendation system structures a report that includes: (i) a narrative summary of the incident, (ii) evidence traces from each agent involved, and (iii) actionable recommendations. These recommendations are generated using a large language model (LLM), which is powered by a structured summary of the correlated events. In this architecture, the LLaMA 3.3-70B model, accessible via the Groq API, is used to synthesize clear and contextualized threat narratives. The LLM is guided by carefully designed models to avoid overgeneralization and ensure consistency in results.

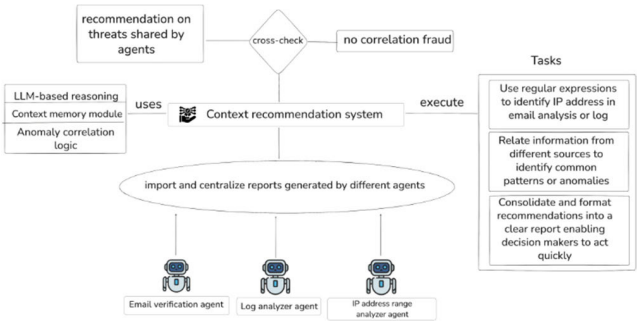


FIGURE 6. The cross-context reasoning architecture.

As shown in Fig. 6, when no significant correlation is found, the system isolates the results and flags them independently. But if a match between different contexts is found, the system generates unified threat intelligence, enriched with human-readable information and priority indicators. These final outputs provide cybersecurity teams with not only raw indicators, but also contextual understanding, accelerating informed decision-making and incident response.

With this systematic consolidation and improved interpretability, the contextual recommendation system moves from a disparate analytics agent architecture to an orchestrated cybersecurity platform that supports real-time threat assessment from multiple, explainable sources.

IV. SYSTEM IMPLEMENTATION AND INTEGRATION

In the following section, we present the implementation resources that support our modular cybersecurity system. This includes the LLMs used for task-specific language processing, the cybersecurity tools integrated into each analytical pipeline, and the datasets employed for validation and performance evaluation. The system executes clearly defined workflows within each agent, ensuring reproducibility, traceability, and operational clarity. Together, these components enable robust, context-aware threat detection across diverse data sources, while maintaining transparency and ease of integration into security operations.

A. LARGE LANGUAGE MODELS

Each analytical agent in our architecture incorporates a LLM to perform domain-specific semantic analysis based on a well-defined and deterministic workflow. The system uses LLaMA 3.3-70B, provided via the Groq API, selected for its exceptional performance in terms of instruction following, long-context understanding, and structured output generation. Each LLM operates strictly within a pre-scripted pipeline, receiving constructed prompts and returning standardized outputs based on its designated role. This ensures control and interpretability across a variety of tasks in the field of cybersecurity analysis.

To improve the accuracy and contextual grounding of LLM responses as shown in Fig.7, retrieval-augmented generation (RAG) is used in important modules like email verification and IP range analysis agents. These agents dynamically retrieve relevant records from external sources, such as phishing repositories or vulnerability databases like the NVD, and provide them as context to the LLM, thus enriching its results without granting it decision-making autonomy. The Chain-of-Thought prompting method is used across all agents, guiding the LLM through a step-by-step analytical reasoning process appropriate for each task. For example, when analyzing log sequences, the model is tasked with extracting relevant timestamps, linking them to potential threat signatures, and reconstructing the event in a narrative form, always following the sequence defined in the workflow.

The prompts are designed to specify the input data, expected formats, and contextual variables specific to the mission of each agent. This approach enhances the reliability of the results and prevents any deviation from the assigned workflow. In parallel, the email verification agent uses semantic similarity via FAISS-based vector matching, where incoming emails are transformed into embeddings and compared to a cleaned index of legitimate and fraudulent messages. This allows the LLM to contextualize new

messages based on their proximity to known patterns, without having to deduce intent or engage in generative reasoning.

By design, this system combines the expressive capabilities of LLMs with a closely structured processing pipeline, ensuring both interpretability and operational security. The result is a scalable and robust cybersecurity system that leverages LLMs as task-specific processors integrated into explainable decision pipeline.

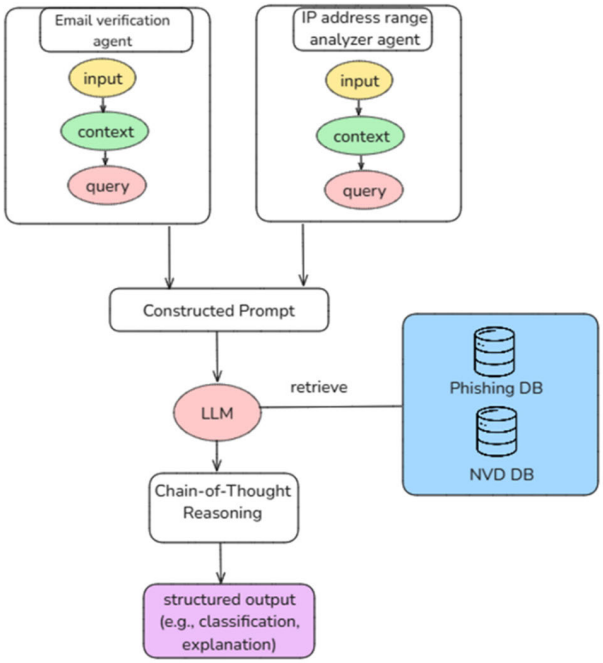


FIGURE 7. RAG architecture.

B. CYBERSECURITY TOOLS

The system architecture also includes a series of cybersecurity tools and utilities to facilitate data collection, pre-processing and scanning, as well as low-level pattern detection. These tools as shown on Table 2 work in parallel with the LLMs to guarantee both technical accuracy and semantic comprehension.

C. DATASETS

In order to evaluate and compare the proposed multi-agent cybersecurity framework, a combination of synthetic and publicly available datasets was selected. These datasets were chosen to ensure realistic coverage of adversarial behavior in email, log and network environments. Each agent was developed and validated using data adapted to its respective domain, to guarantee classification accuracy and semantic interpretability.

Although the SpamAssassin and CIC-IDS2017 datasets are considered relatively old, they continue to serve as standard references in recent studies [56], [57]. Mondragon et al. [56] compared modern IDS datasets and included CIC-IDS2017 due to its wealth of attack patterns and labeled flows. Similarly, a 2024 study [57] confirms its continued relevance for both model evaluation and reproducibility. Their use allows for comparison with previous research while supporting the generalization testing of our agents under known threat conditions.

TABLE 2. Cybersecurity tools.

TOOL	PURPOSE	USED BY
FAISS	Vector similarity search for email classification	Email Verification Agent
ELK Stack	Log ingestion, indexing, and query processing	Log Analyzer Agent
Suricata	Intrusion detection through rule-based pattern recognition	Log Analyzer Agent
Nmap	Active scanning of IP ranges, port detection, and OS fingerprinting	IP Range Analyzer Agent
NVD API	Extraction of vulnerability data and CVE details	IP Range Analyzer Agent
ipaddress (Python module)	Parsing and validating CIDR notations and IP inputs	IP Range Analyzer Agent
tldextract	Extracting base domains from email URLs	Email Verification Agent
DNS.resolver	Performing DNS lookups for domain validation	Email Verification Agent
Regular Expressions	Identifying suspicious strings, keywords, or attack signatures	All agents
TextBlob	Performing sentiment analysis on email content	Email Verification Agent

1) SPAM ASSASSIN PUBLIC EMAIL CORPUS

The SpamAssassin Public Corpus is a widely used email dataset that includes both unwanted messages (spam) and legitimate messages (ham). The collection is a fundamental resource for evaluating spam detection and phishing detection models. In the current setting, this dataset was used to build a semantic index based on FAISS for the email agent. The dataset also served as a reference for LLM prompt generation and classification benchmarking. It contains approximately 6,000 messages, each labeled and structured for research purposes. Both the vector search component and the contextual analysis modules of the email verification agent were powered by this resource [12].

Although the SpamAssassin dataset is relatively old, it remains a standard reference in research on email spam detection. Its availability, clear labeling, and widespread adoption in previous studies make it valuable for reproducibility and comparative evaluation. Although there are more recent email datasets available, these are often not publicly accessible or do not provide sufficient annotation quality to enable multi-agent semantic correlation.

2) CIC-IDS 2017 DATASET

The CIC-IDS 2017 dataset, developed by the Canadian Cybersecurity Institute, provides a comprehensive

compilation of logs and packet captures from various cyber-attack simulations. It includes real network traffic and behavioral anomalies such as brute force attacks, intrusions, botnet activity, and DDoS attacks. This dataset was used to evaluate the ability of the log analyzer agent to identify suspicious behavior and generate natural language interpretations using LLMs. The dataset validated the detection of temporal anomalies, sequence extraction, and cross-domain event mapping. The format of the dataset and the coverage of attacks make it particularly suitable for log-based machine learning systems [13].

Despite its age, CIC-IDS 2017 continues to be one of the most comprehensive labeled datasets for intrusion detection, covering a wide range of attack types. Its use facilitates compatibility with previous research and enables baseline performance validation. Nevertheless, we recognize that Internet protocols and attack vectors have evolved since its publication.

3) SYNTHETIC PHISHING EMAILS GENERATED BY LLMS

To supplement real data from emails and simulate advanced phishing techniques, a customized dataset was generated using language models adapted to the instructions. These models produced a diverse set of phishing messages that reflect modern adversary strategies, including psychological manipulation, domain spoofing and multilingual targeting. The synthetic dataset contains approximately 2,000 examples and was specifically designed to test the robustness of the semantic and contextual reasoning capabilities of the email agent. It was also used to evaluate the ability of the model to interpret new threats and return explainable classifications under ambiguous or unfavorable conditions.

The synthetic dataset was generated to simulate advanced phishing attempts and adversaries that are not listed in public datasets. Although it follows realistic construction rules and internal annotations, it is not made public due to confidentiality constraints and its validation status.

4) SIMULATED IP SCANNING AND VULNERABILITY ASSESSMENT

A controlled test environment was established to evaluate the IP range analyzer agent, using virtual machines configured with obsolete services and known vulnerabilities. Each host was scanned using Nmap, and the corresponding service indicators were mapped to vulnerability data extracted from the national vulnerability database. This configuration validated the ability of the agent to scan, extract relevant service metadata, retrieve accurate CVEs, and summarize risk exposure using semantic prompts. It also served as a baseline for evaluating the quality of the summaries generated by the LLM and the results of the decision support.

5) DATA PREPROCESSING AND USAGE

Each dataset was preprocessed according to its structure and the requirements of the corresponding agents. For the SpamAssassin corpus, email headers and body texts were parsed and tokenized using NLTK, and spam labels were retained to simulate detection signals for the Email Verification Agent. HTML tags and malformed characters were removed to ensure clean semantic parsing by the LLM.

For the CIC-IDS2017 dataset, we extracted flow-based features including source/destination IPs, ports, timestamps, and attack labels. These were used by the Log Analyzer Agent to detect anomalies and later by the Correlation Agent to match IPs and timestamps with email events and scan metadata. The data was filtered to retain only malicious and ambiguous sessions relevant to CVE-based correlation.

The custom dataset of 60 tasks was built by merging synthetic email messages (including obfuscated indicators), Nmap scan outputs (for vulnerable ports), and log sequences (simulated from CIC flow data). Each task was designed to test specific reasoning patterns: single-agent detection, multi-agent inconsistency resolution, false positive filtering, and semantic explanation generation.

All datasets were unified under a JSON-based task format and indexed to be processed sequentially during evaluation. Preprocessing scripts and task generation pipelines will be made available upon request.

The following Table 3 summarizes the role of each dataset within the multi-agent system:

The implementation of the Log Analyzer agent is based on recent research on system log analysis and anomaly detection using deep learning. Le and Zhang [3] provided a comprehensive review of deep learning techniques applied to system logs, highlighting both their effectiveness and limitations in real-world environments. Their study focused on the challenges of generalization, false positives, and log diversity, which our agent addresses through modular reasoning and correlation layers. In addition, Xie et al. [4] presented LogGD, a graph neural network-based approach capable of capturing structural dependencies in log sequences. Their work demonstrated the importance of integrating contextual relationships between logs for accurate anomaly detection. These results confirm our architectural choices, which favor hybrid symbolic-neural reasoning over purely statistical classifiers for anomaly detection in logs.

D. LLM INTEGRATION STRATEGY

In our multi-agent cybersecurity system, LLMs are not used as autonomous cognitive entities, but as structured semantic modules integrated into predefined analytical workflows. As shown in Fig.8, each agent operates according to clearly scripted instructions, and the role of the LLM is limited to extracting, interpreting, and generating semantically enriched results to support these workflows. This architectural choice ensures maximum interpretability, reproducibility and operational reliability.

Each analytical agent dynamically generates prompts that are precisely adapted to the context of the data it is processing. These prompts include a clear definition of the task, a detailed structure of the inputs and the expected formatting of the outputs. For example, in the email verification agent,

prompts include information about the sender, message headers, textual content and embedded URLs. The LLM then evaluates these components to provide structured assessments of phishing risk, urgency or identity fraud attempts. Similarly, the IP range analyzer agent formulates prompts that contextualize Nmap scan results and vulnerability metadata to produce semantic summaries describing host exposure and potential exploitability.

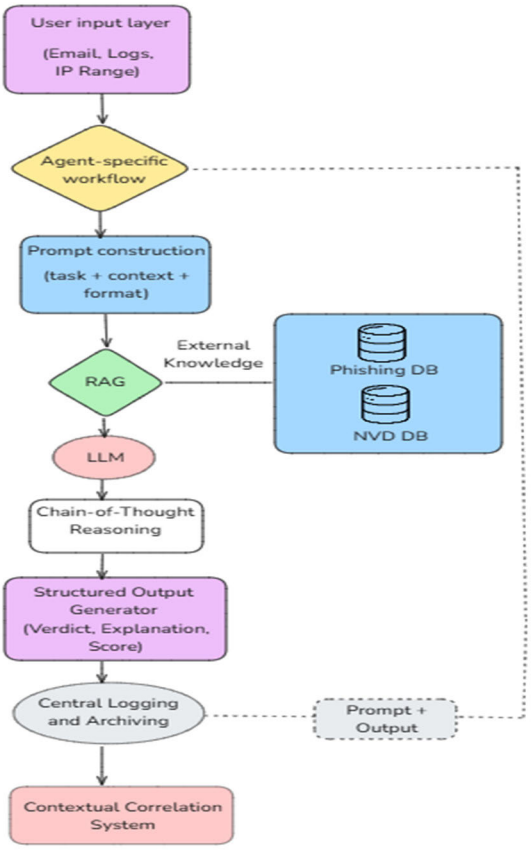


FIGURE 8. LLM integration pipeline within the multi-agent workflow.

TABLE 3. Dataset utilization overview.

DATASET	DOMAIN	ASSOCIATED AGENT	PURPOSE
SpamAssassin Public Corpus	Email	Email Verification Agent	Semantic similarity search, fraud detection training, prompt grounding
CIC-IDS 2017	Email	Email Verification Agent	Anomaly detection, event interpretation, behavior- based reasoning
Synthetic Phishing Emails	Email	Email Verification Agent	Model robustness evaluation, adversarial input testing
Simulated IP Scans	Network	IP Range Analyzer Agent	Vulnerability extraction, Nmap validation, semantic risk summary generation

This system relies heavily on chain-of-thought prompts across all agents. This technique guides the LLM through intermediate reasoning steps to improve transparency by breaking down complex assessments into traceable, human-readable logic. The interpretive value of this approach is amplified by its consistency across all detection domains. In addition, retrieval-augmented generation (RAG) is selectively applied in components such as email and IP agents. RAG enables the system to dynamically integrate external knowledge sources such as phishing datasets or CVE repositories into the prompt context, significantly improving topical relevance and domain alignment.

All LLM inputs and outputs, including associated prompts, are systematically recorded and archived. This ensures complete decision traceability and allows analysts to reconstruct and audit any step of the semantic processing. These records also serve as a training ground for red teaming exercises and for refining prompt design over time, without requiring retraining or adjustment of the model itself.

The strategy for integrating LLMs into the architecture therefore emphasizes modularity, semantic clarity and repeatability. By integrating LLMs into the system as deterministic, prompt-driven components instead of granting them autonomy to reason, we create a secure and scalable semantic processing layer. This integration supports the broader system goals of interpretable detection, workflow control and cross-domain threat correlation, while ensuring alignment with operational security standards and analyst expectations.

V. DISCUSSION

This section presents a detailed evaluation of the performance and operational viability of the proposed system. The discussion summarizes the evaluation results from both isolated agent tests and full pipeline integration, providing quantitative insights into detection accuracy, semantic interpretability and cross-domain correlation effectiveness. A comparative analysis with existing peer-reviewed cybersecurity solutions is also included, highlighting the relative strengths of the system in terms of modularity, semantic clarity and workflow efficiency. Together, these results establish the practical suitability of the system and provide important insights into its scalability, explainability and real-world applicability.

A. EVALUATION SYSTEM

In order to evaluate the effectiveness and robustness of the proposed modular cybersecurity system, we conducted a comprehensive assessment in a controlled experimental environment [13]. The evaluation was carried out in two stages: independent agent validation and integrated pipeline testing.

Each agent was validated individually using datasets tailored to its operational scope. The Email Verification Agent was assessed on the SpamAssassin corpus [15], widely adopted in both classic and recent phishing detection research [7]. The Log Analyzer Agent was tested using system log sequences derived from the CIC-IDS2017 dataset [12], which remains a widely used benchmark for

TABLE 4. Performance metrics and their mathematical definitions.

Metric Name	Mathematical Definition	Component Application
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Measures correct predictions in all agents
Precision	$\frac{TP}{TP+FP}$	Used in email and IP agents to reduce false alerts
Recall	$\frac{TP}{TP+FN}$	Critical for the Log Analyzer to capture rare events
F1-Score	$2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$	Balances recall and precision
False Positive Rate	$\frac{FP}{FP+TN}$	Used to assess alert quality across agents
Correlation Precision	$\frac{Correct\ Correlations}{Total\ Correlation\ Attempts}$ inputs	Used by the Contextual Recommendation System
Analyst Response Time Reduction	$\frac{Tbaseline\text{-LLM}}{Tbaseline}$	Measures impact of explainability features

log-based intrusion scenarios [5]. For the IP Range Analyzer Agent, synthetic IP scanning scenarios were constructed using realistic configurations and matched against CVE data from the National Vulnerability Database [21].

These datasets collectively represent a broad spectrum of threats—including phishing campaigns, abnormal system logs, and host/network enumeration attempts [19], [20], [22]. Following the isolated validation of each agent, we evaluated the complete workflow to measure inter-agent communication, multi-domain correlation, and overall system coherence [6], [11], [16].

Performance was measured using standard metrics from artificial intelligence and cybersecurity, as shown in Table 4 [8], [17]. These metrics included accuracy (correct classification rate), precision (rate of true positives among predicted positives), recall (rate of true positives among actual positives), F1 score (harmonic mean of accuracy and recall), false positive rate (FPR), correlation accuracy (accuracy of the link between inter-agent events), and reduction in analyst response time (time saved thanks to the explanation and synthesis provided by LLMs) [9], [18], [35]. Each indicator was formulated mathematically to reflect its diagnostic usefulness across all components of the system [2], [24], [36].

B. RESULTS

In accordance with the evaluation methodology described in Section A, the results are presented from three interdependent perspectives: the individual performance of each agent, the collective behavior of the system in full integration, and its ranking relative to peer-reviewed cybersecurity architectures [7], [18]. This layered analysis reveals not only the detection and semantic capabilities of the individual components of the system, but also their combined ability to provide reliable and contextual threat descriptions at scale [5], [24], [31].

Each agent, specialized respectively in email verification, log analysis, and IP range analysis, was tested using a domain-specific dataset reflecting realistic threat conditions [12], [13], [14]. As shown in Table 5, all agents achieved good performance in terms of classification accuracy, with values ranging from 91.8% to 94.1%. The semantic interpretability of the system is equally impressive: over 90% of the explanations generated by LLMs were judged to be both technically valid and useful in practice [2], [25], [39]. The IP range analyzer achieved the highest success rate for explanation, at 93.2%, closely followed by the email verification module [15], [30]. The log analyzer, while significantly lower in terms of explanation rates, clearly benefited from the use of the thought chain, which enabled the LLM to express sequential behavior and complex anomaly chains in a readable and informative manner [9], [21], [22]. These observations validate the effectiveness of integrating LLMs into a deterministic workflow, where the language model acts as a modular and contextual summarizer for the task at hand [4], [16], [29].

Once all components were integrated into the unified architecture, the system demonstrated its capability to synthesize disparate threat signals from email content, system logs and network exposure analysis. The contextual recommendation system played a central role in correlating indicators across domains, identifying common temporal and behavioral patterns, and formulating a single narrative for each threat event. This cross-domain synthesis capability was evaluated using system-wide performance metrics, as shown in Table 6. The architecture achieved an overall detection accuracy of 93.6% and a correlation accuracy of 87.0%, indicating effective linking between events from different agent pipelines. In addition, false positive alerts were reduced by 41.3% compared to a traditional SIEM reference, and triage time for analysts decreased by 38.5% on average due to clear, contextual explanations accompanying each automated decision. Analysts rated the interpretability and relevance of the results with an average confidence score of 4.6 on a 5-point scale, highlighting the operational viability of the system.

To contextualize the performance of our proposed system within the larger cybersecurity research landscape, we conducted an in-depth comparison with twelve peer-reviewed systems that incorporate LLMs, multi-agent designs, or hybrid detection architectures [17], [18], [27]. These systems represent a wide range of approaches, including semantic enrichment, phishing detection, OSINT integration and IoT threat analysis [10], [24], [35].

As shown in Table 6, our system demonstrates superior performance in terms of accuracy and F1 score. With a classification accuracy of 93.6% and an F1 score of 0.94, it outperforms the best previously reported scores among

the compared systems. It should be noted that while several studies, such as those by Koide et al. [3] and Zhang et al. [27], have demonstrated good results with variants of GPT-4, they were generally limited to specific domains (e.g., phishing websites, synthetic benchmarks) and did not allow for the integration of multiple data types. Systems such as those by Kaluzhnaya et al. [16] and Pratama et al. [15] adopted multi-agent approaches, but they did not include inter-contextual correlation or semantic normalization of results [5], [11].

TABLE 5. Agent-level performance metrics.

AGENT	ACCURACY (%)	FALSE POSITIVE RATE (%)	EXPLANATION SUCCESS(%)
Log Analyzer Agent	91.8	8.7	89.5
IP Range Analyzer Agent	92.7	5.4	93.2
Email Verification Agent	94.1	6.2	91.4

TABLE 6. Full pipeline evaluation.

METRIC	RESULT
System-Wide Accuracy	93.6%
Threat Correlation Precision	87.0%
False Positive Reduction (compared to baseline SIEM)	41.3%
Average Analyst Triage Time Reduction	38.5%
Mean Analyst Trust Score (on a 1 to 5 scale)	4.6

In addition, our system stands out for its modular design, structured workflow and transparent semantic layer, which enable consistent interpretability across logs, emails and network analysis [2], [25], [31]. This integrated configuration not only achieves high accuracy but also increases analyst confidence and improves response time, two factors that are essential for operational deployment [24], [36], [39].

Performance test results confirm the benefits of our deterministic, workflow-driven model, which offers a balance between automation and human readability [28], [37], [38]. By structuring tasks into dedicated agents and leveraging LLMs through retrieval-augmented generation (RAG) and thought-chain prompts, the system achieves high accuracy without sacrificing visibility or scalability [4], [6], [29].

To obtain the results presented in Tables 5 and 6, a two-step evaluation protocol was followed. In the first step, the system was tested using labeled samples from the three datasets (SpamAssassin, CIC-IDS 2017, and synthetic phishing emails). Each agent processed its corresponding domain (emails, logs, IP address ranges) and the results were collected to calculate accuracy. Measures such as true positive rate, false positive rate, and accuracy were calculated for each agent and then aggregated.

In a second step, 60 anonymized tasks were created from a mix of the datasets and submitted to the entire multi-agent pipeline. For each task, the final recommendation report was evaluated by three human analysts who were unaware of the ground truth. They rated each report based on its clarity, reliability, and usefulness, and indicated the time saved compared to a manual inspection.

Table 5 shows the classification accuracy of the system for all entries. Table 6 summarizes the qualitative comments from the analysts for all scenarios.

In order to enhance the interpretability and comparative analysis of the results. Fig.9 presents a heat map of performance metrics at the agent level, including classification accuracy, false positive rate (FPR), and explanation success. The visualization highlights the high accuracy of the email verification agent, the semantic power of the IP range analyzer, and the relatively higher FPR of the log analyzer. This visual summary provides a clear comparative view of each agent's performance against key evaluation criteria.

Fig.10 shows the overall improvements achieved by the proposed system compared to a basic SIEM reference. The architecture demonstrates a significant 41.3% reduction in false positives and a 38.5% reduction in analyst triage time, confirming the operational effectiveness and efficiency of the system in practical deployment environments.

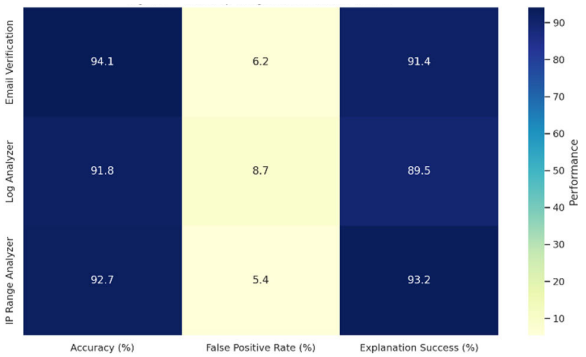


FIGURE 9. Heatmap of agent-level performance metrics.

To better illustrate the capabilities of the proposed system in terms of threat detection and multi-agent collaboration, we provide two additional graphs based on 60 evaluation tasks. These graphs offer a dynamic view of the internal

TABLE 7. Benchmarking with comparable multi-agent or LLM-Based Cybersecurity systems.

Study	Focus Area	LLM Purpose	Explainability	Agent-based	Correlation Capability	Data Domains	Adaptability	Operational Maturity
Afane et al [1]	Simulated Phishing Generation	Offensive (simulate adversary)	None	No	No	Email only	Static simulation	Conceptual only
Koide et al [3]	Phishing Page Detection	Classify with CoT prompts	Human-readable prompts	No	No	Web phishing URLs	Limited to visual/text	Ready for online testing
Nasution et al. [45]	Benchmark LLMs on URL Phishing	Prompt-based detection	Binary outputs only	No	No	Text-based phishing URLs	21 LLMs tested	Lab evaluation only
Li et al. [47]	Brand Spoofing Detection	Text–Visual Fusion with Knowledge Graph	Narrative + visual reasoning	Hybrid fusion	Entity alignment only	Full webpage content	Works with/without logos	Robust anti-phishing defense
Wang et al [46]	Phishing URL Analysis	Pretrained Transformer (PhishBERT)	No semantic reasoning	No	No	URLs (2M+ dataset)	Static BERT-style	High-throughput URL filter
Balasubramanian et al. [48]	Log-based Anomaly + Chatbot	Summarize + reason with GPT-3	Chat-based response rationale	No	No	Logs (auth/system)	Dialog-driven analysis	Conversational SIEM layer
Karlsen et al. [49]	Log Security Benchmarking	Fine-tuned Transformers	Attention & attribution	No	No	Diverse log datasets	Transferable models	Deployable with logs
ZeMicheal et al [50]	CVE Validation + Triaging	LLM for contextual CVE analysis	Justified per CVE	Yes (planner–executo)	Local CVE checks only	Container security (Docker)	System-aware	Analyst-grade automation
Koucham et al. [51]	ICS Multi-Domain Correlation	No LLM	Context-aware alert fusion	Multi-source sensors	Time-based + process state	Cyber + physical ICS data	Domain-specific only	Real-world ICS tested
Our Work	**Multi-Layer Cybersecurity (Logs, Email, IP)**	**Semantic Detection + Justification via LLaMA + RAG**	**CoT + Full Narrative Explanation**	**Modular Agents + Central Correlator**	**Inter-domain event fusion**	**Email, Logs, IP, CVE**	**Prompt-based, highly extendable**	**Operational & explainable across domains**

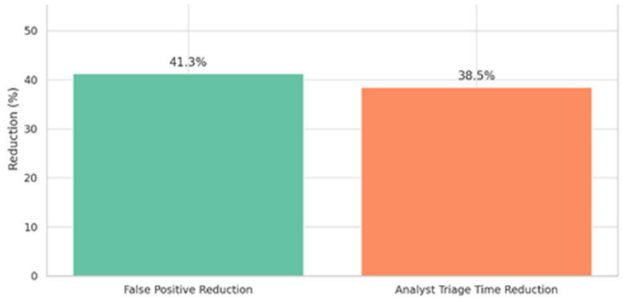


FIGURE 10. System-wide improvements compared to baseline SIEM.

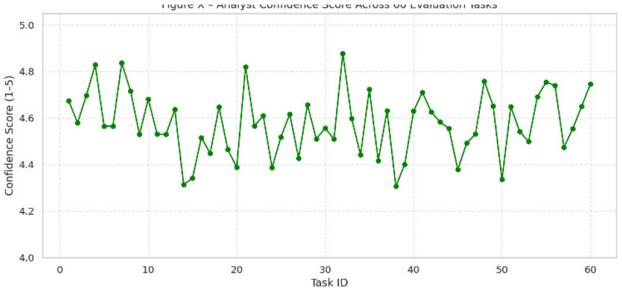


FIGURE 11. Confidence score of the analyst across 60 evaluation.

reasoning and correlation behavior of the system, beyond static metrics.

The first graph Fig.11 illustrates the analysts confidence score for each task. This score, rated on a scale of 1 to 5, was estimated based on the clarity, consistency, and relevance of the explanations generated by the system during correlation. As shown in the graph, most tasks received confidence scores

above 4.5, highlighting the ability of the system to generate reliable and contextually consistent results.

The second graph Fig.12 shows the number of agents (email, log, IP) involved in each correlated decision. This graph confirms that in most cases, two or more agents jointly contributed to a final decision regarding the threat, demonstrating the strength of the cooperative correlation mechanism. It also reflects the complementary nature of the agents' roles in covering different aspects of the cyber attack surface.

This visual information helps reinforce the effectiveness and reliability of our multi-agent architecture for real-world cybersecurity applications.

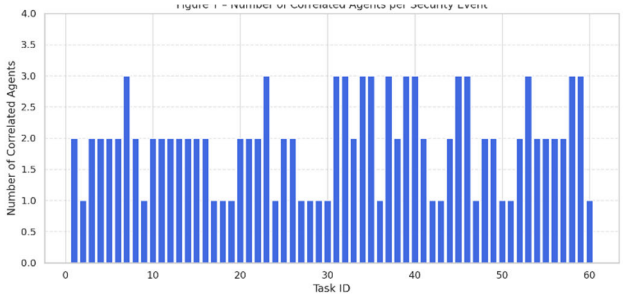


FIGURE 12. Number of correlated agents per security event.

VI. CONCLUSION AND FUTURE WORK

This study presented a multi-agent cybersecurity architecture designed to address the growing demand for semantically enriched and explainable threat detection. The proposed system leverages LLMs not as autonomous decision makers, but as structured processing units operating within clearly defined workflows. Agents focus on specific domains (email verification, log analysis and IP range inspection), and their results are centrally synthesized by a contextual recommendation system that performs cross-domain correlation. This architecture enables the system to go beyond static signature-based alerts and provide a more coherent and traceable interpretation of evolving cyber threats.

An extensive evaluation using public and adversarial datasets confirmed the robustness and usefulness of the system. With a system-wide detection accuracy of 93.6%, the platform demonstrated not only strong classification capabilities, but also high-quality semantic explanations, reduced false positives and improved sorting efficiency for human analysts. However, performance metrics are not the only thing that matter. The architecture has proven its ability to deliver structured and transparent results, which is essential in security operations where traceability, compliance and human oversight remain non-negotiable.

One of the major of the system strengths lies in its modular scalability. Each agent operates independently in its domain of analysis while contributing to a shared correlation layer with standardized results. This division of work allows for gradual evolution of the system without compromising the cohesion or consistency of results. Thus, the architecture is not only a detection tool, but a coordinated decision support system adapted to operational environments that require both interpretability and actionability.

Future work will aim to extend the capabilities of the framework without compromising its design principles. One direction for development is to integrate multimodal inputs, such as attachments, embedded links, and visual payloads, which would extend the detection capability in phishing and advanced social engineering contexts. Another path is to improve real-time responsiveness by adapting the log analysis agent to handle continuous data streams. Federated deployment models will also be explored to address scenarios involving distributed infrastructure and local privacy constraints. Finally, integrating lighter LLM variants and refining rapid engineering strategies could facilitate deployment on edge systems or in low-bandwidth environments, while preserving semantic clarity and result traceability.

Despite the encouraging results and solid architecture design, this study has some limitations and threats to validity that need to be considered for practical deployment and generalization.

First, the system relies on third-party tools and APIs (e.g., Nmap, Groq), which can cause latency issues, especially when deployed in high-throughput or real-time environments. These dependencies can also become single points of failure if services are interrupted or limited in throughput.

Second, the integration of LLM for explanation generation, while powerful, introduces computational cost and potential unpredictability depending on prompt construction or model updates. Although prompts are currently modeled and agents operate in a modular fashion, consistency of explainability across updates remains a challenge.

Third, the use of legacy data sets, while justified for comparative analysis, limits performance evaluation in the context of emerging and constantly evolving threat scenarios.

To mitigate these limitations, the architecture has been purposefully designed with modularity, replaceability, and fallback in mind. For example, external LLMs can be replaced with local models in restricted environments. Future iterations will also explore asynchronous task execution and resilience strategies to reduce latency and improve robustness in production environments.

This research contributes to the development of a deployable, transparent and adaptable cybersecurity system. The research demonstrates that when structured workflows and LLM reasoning are properly combined, organizations can achieve both highly accurate threat detection and operational clarity without resorting to anecdotal information or opaque automation. The architecture provides a foundation for further innovation in the field of explainable, human-centric cyber defense.

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